

SCOTTSDALE, AZ, USA

WMO: 722789

Lat: 33.623N

Lon: 111.911W

Elev: 449

StdP: 96.05

Time Zone: -7.00 (W07)

Period: 01-19

WBAN: 03192

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.3	3.8	-16.7	0.9	17.2	-14.1	1.2	18.5	7.5	13.1	6.0	14.0	0.5	50	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	42.6	20.3	41.5	20.2	40.4	20.0	23.5	33.0	23.1	33.2	22.7	33.4	3.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	17.0	26.8	20.3	15.9	27.9	19.2	14.7	29.2	72.4	32.7	70.8	33.0	69.1	33.0	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.1	5.6	4.8	DB	-0.6	44.8	1.4	1.2	-1.7	45.7	-2.5	46.4	-3.3	47.1	-4.4	48.0
			WB	-3.2	24.5	1.5	0.5	-4.3	24.8	-5.2	25.1	-6.0	25.4	-7.2	25.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	12.9	14.2	18.1	21.7	26.1	32.1	34.2	33.4	30.3	23.9	17.6	12.1		
	DBStd	8.45	3.13	3.49	3.68	3.30	3.56	2.70	2.28	2.37	2.87	3.45	3.73	3.16		
	HDD10.0	35	11	6	1	0	0	0	0	0	0	0	1	16		
	HDD18.3	603	169	121	49	9	1	0	0	0	0	4	57	193		
	CDD10.0	4812	100	122	252	351	499	662	752	724	610	431	228	82		
	CDD18.3	2339	1	3	42	110	242	412	493	466	360	176	34	1		
	CDH23.3	35583	19	81	551	1321	3327	6828	8250	7588	5303	1954	351	9		
	CDH26.7	22361	1	7	158	528	1837	4626	5802	5167	3257	893	84	0		
(14) Wind	WSAvg	1.6	1.1	1.3	1.6	2.0	2.1	1.9	2.2	1.9	1.7	1.4	1.2	1.1		
(15) Precipitation	PrecAvg	311	42	37	37	13	6	3	28	43	25	21	20	36		
	PrecMax	619	218	141	131	47	53	17	81	118	109	142	90	122		
	PrecMin	137	1	0	1	0	0	1	2	11	0	1	1	2		
	PrecStd	111	48	38	30	12	11	3	21	24	23	26	19	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	27.7	32.8	35.4	39.7	44.1	43.9	43.4	41.2	36.8	31.4	24.5		
		MCWB	11.0	12.2	14.4	14.9	17.2	18.8	20.8	20.8	20.3	17.5	14.9	11.5		
	2%	DB	23.4	26.0	30.2	33.4	37.7	42.2	42.6	42.1	39.6	34.6	28.9	22.6		
		MCWB	10.6	11.7	13.5	14.3	16.7	18.5	20.9	21.1	19.9	16.9	13.8	11.1		
	5%	DB	21.8	23.9	28.3	31.6	36.3	41.0	41.6	40.9	38.4	33.0	27.2	20.8		
		MCWB	10.2	10.8	12.7	13.7	15.9	18.5	20.8	20.9	19.5	16.4	13.4	10.5		
	10%	DB	19.9	21.9	26.6	29.8	34.6	39.6	40.3	39.4	37.2	31.6	25.3	18.8		
		MCWB	9.6	10.3	12.1	13.0	15.4	18.0	20.8	20.9	19.4	16.0	12.6	9.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	14.5	15.4	16.3	19.2	22.0	24.1	24.1	23.4	21.4	16.8	14.6		
		MCDB	18.6	20.8	28.2	31.3	33.0	35.8	32.8	32.9	31.2	26.7	24.0	18.9		
	2%	WB	12.8	13.4	14.5	15.2	18.0	21.0	23.6	23.6	22.7	19.8	15.4	13.0		
		MCDB	17.7	19.8	27.2	30.6	33.2	36.2	33.1	33.0	31.5	29.0	24.3	19.1		
	5%	WB	11.8	12.4	13.6	14.4	16.8	20.1	23.1	23.2	22.0	18.5	14.5	11.8		
		MCDB	18.0	19.5	25.6	28.9	34.0	36.8	33.4	33.2	31.9	28.5	24.1	17.5		
	10%	WB	10.8	11.5	12.8	13.7	15.9	19.1	22.8	22.8	21.5	17.4	13.6	10.7		
		MCDB	17.7	19.5	24.1	27.8	33.2	36.9	33.8	33.4	32.1	28.4	23.5	16.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.1	13.2	13.1	13.2	13.6	11.2	11.1	11.7	12.5	12.5	11.5		
		MCDBR	15.2	15.6	15.8	15.6	15.2	14.7	12.3	12.6	12.9	14.2	14.8	14.3		
	5% WB	MCWBR	6.5	6.2	5.1	5.1	4.5	4.0	3.2	3.0	3.5	4.5	5.5	6.2		
		MCWBR	11.0	11.5	14.2	13.7	13.8	12.0	10.5	10.3	9.6	10.3	11.8	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.286	0.313	0.322	0.331	0.341	0.402	0.389	0.344	0.321	0.289	0.280		
	taud		2.574	2.570	2.472	2.435	2.442	2.414	2.371	2.396	2.496	2.510	2.554	2.579		
	Ebn at Noon		948	975	965	964	952	938	879	888	920	920	927	928		
	Edh at Noon		78	87	105	114	114	117	122	117	101	92	79	74		
(44) All-Sky Solar Radiation	RadAvg		3.35	4.30	5.80	7.23	8.12	8.45	7.34	6.77	6.08	4.89	3.73	3.03		
	RadStd		0.26	0.35	0.29	0.22	0.30	0.32	0.34	0.23	0.23	0.30	0.17	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+98		
(47)	Regional (15 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	-81	+99	+34		

Nomenclature: See separate page